SECTION 1

Time—30 minutes

40 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. SOOTHE: (A) praise (B) promise (C) provoke (D) join together (E) conceal quickly

2. DECELERATION: (A) increased speed (B) regular occurrence (C) additional cost (D) different approach (E) greater productivity

3. LINGER: (A) inflate (B) detest (C) attempt to win (D) thrust in deeper (E) hasten to depart

4. INCONSPICUOUS: (A) blameworthy (B) glaring (C) possible (D) beneficial (E) desperate

5. FRACTURE: (A) knit (B) disencumber (C) entangle (D) render inactive (E) pull taut

6. ACCESSORY:
(A) logical argument
(B) persistent questioner
(C) inconclusive evidence
(D) primary component
(E) distinguished predecessor

7. DISPASSIONATE: (A) isolated (B) prejudiced (C) unfortunate (D) irreconcilable (E) disgraceful

8. SUPERCILIOUS: (A) fatal (B) acute (C) extreme (D) servile (E) precocious

9. DESECRATE: (A) refurbish (B) construct (C) moisten (D) rejoice (E) hallow

10. TANTAMOUNT:
(A) unclear in meaning
(B) impractical in design
(C) undesirable in effect
(D) immeasurable in size
(E) unlike in significance

GO ON TO THE NEXT PAGE

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1

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ---, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate
(C) sophisticated. .amateur (D) risqué. .crude
(E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

11. Lee knows the jungle and loves it, having the explorer's ---- to ---- discomfort for the sake of knowledge gained through the senses.

(A) zeal. .lessen
(B) resolve. .display
(C) willingness. .endure
(D) disinclination. .discover
(E) fervor. .overestimate

12. Some scientists argue against ---- any microbe, suggesting that even the smallpox virus, which is now nearly ----, might be preserved in a kind of microbial museum.

(A) conserving. .isolated
(B) exterminating. .extinct
(C) eradicating. .immunized
(D) fertilizing. .virulent
(E) infecting. .controlled

13. People who insist that some particular injustice is not their responsibility sooner or later become unable to ---- any injustice.

(A) condone (B) excuse (C) permit
(D) ignore (E) resent

14. The Modernists were people of the strongest ---- propensities: they gloried in their refusal to ---- bourgeois sentiments and relationships.

(A) altruistic. .wink at
(B) radical. .cast off
(C) puritanical. .preside over
(D) antisocial. .traffic in
(E) conservative. .abide by

15. The establishment of a center here has been held up by insistent fears that local ---- would be undermined by state control.

(A) autonomy (B) officials (C) buildings
(D) recreation (E) funds

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

16. PADDLE : CANOE :: (A) runner : sled
(B) pedal : bicycle (C) hangar : airplane
(D) track : train (E) dock : boat

17. TELESCOPE : ASTRONOMER ::
(A) binoculars : oculist
(B) stethoscope : chemist
(C) periscope : aviator
(D) horoscope : astrologer
(E) microscope : biologist

18. SHOPLIFTER : MERCHANT ::
(A) burglar : homeowner (B) pickpocket : thief
(C) defendant : juror (D) colleague : employer
(E) browser : customer

19. MIST : DAMPNESS :: (A) ice : slipperiness
(B) hail : heat (C) snow : winter
(D) climate : humidity (E) wind : thaw

20. BRAGGART : BOAST ::
(A) censor : obscenity (B) heckler : compliment
(C) liar : falsehood (D) flatterer : grievance
(E) conspirator : confession

21. ANESTHETIC : PAIN :: (A) sandpaper : friction
(B) alloy : durability (C) fan : ventilation
(D) preservative : decay (E) spark : electricity

22. SUSPICIOUS : DOUBT :: (A) happy : laughter
(B) apprehensive : fear (C) guilty : conscience
(D) angry : violence (E) imaginative : fiction

23. REVERIE : DAYDREAMER :: (A) plot : conspirator
(B) audience : actor (C) sleep : insomniac
(D) crime : detective (E) perfection : bungler

24. MAVERICK : HERD :: (A) boss : workers
(B) dissenter : group (C) duck : flock
(D) hero : army (E) expert : mass

25. EPICURE : FASTIDIOUS ::
(A) ascetic : abstemious (B) stoic : volatile
(C) spartan : extravagant (D) gourmet : greedy
(E) truant : insubordinate

GO ON TO THE NEXT PAGE

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

There was a young girl of nineteen in a small town of north Germany who had a strong bent for research; but when her brother went to the University of Gŏttingen, she, according to the customs of her country, remained at home. Agnes Pockels had observed the streaming of currents when salts were put into solution and, by attaching a float to a balance, had found that salts increased the pull of the surface of the fluid. In other words, she had discovered surface tension. This was in 1881. She did not know whether anyone else had ever observed this phenomenon. Through her brother, she brought her work to the attention of the professor of physics at Gŏttingen. He failed to grasp its significance. For ten years she went on studying the properties of solutions. Then, when the renowned English physicist, Lord Rayleigh, began to publish on this subject, Agnes wrote to him about her work. Lord Rayleigh sent a translation of her letter to the English journal Nature asking that it be published. He wrote that the first part of her letter covered nearly the same ground as his own recent work and that with very "homely appliances" she had arrived at valuable results respecting the behavior of contaminated water surfaces. (Interestingly, this same "homely device" is still used to measure surface tension.) For a few years Lord Rayleigh arranged for the publication of all Agnes Pockels' work in English until, in 1898, her discoveries were accepted for publication in her own language.

26. Which of the following best describes the content of the passage?

(A) An explanation of the means by which surface tension may be observed and measured
(B) An analysis of the events which led to the discovery of surface tension
(C) An evaluation of the significance of Agnes Pockels' discovery of surface tension
(D) An account of Agnes Pockels' discovery of surface tension and how her work became known to the world
(E) An account of Agnes Pockels' efforts to attend a university and of her reaction to the fame her discovery brought

27. It can be inferred that Lord Rayleigh arranged for the publication of Agnes Pockels' work primarily because he wanted to

(A) reveal the crudity of her equipment
(B) challenge the accuracy of her results
(C) show others the major influences on his own work
(D) assume responsibility for having inspired her first efforts
(E) inform other scientists of the significance of her discoveries

28. It can be inferred that a scientist today would most probably describe the device Agnes Pockels used to measure surface tension as

(A) overly complex
(B) clever but inaccurate
(C) still very effective
(D) of historical value only
(E) difficult to reconstruct

GO ON TO THE NEXT PAGE

The reading passages in this test are brief excerpts or adaptations of excerpts from published material. The ideas contained in them do not necessarily represent the opinions of the College Board or Educational Testing Service. To make the text suitable for testing purposes, we may in some cases have altered the style, contents, or point of view of the original.

Environments spoiled by smogs, pesticides, or strip mining are not destroyed thereby but rather become different by evolving in new ecological directions determined by these stresses. We may not like the aesthetic and economic consequences of these alterations, but we can be sure that the disturbed environments will eventually achieve some new kind of biological equilibrium, as has repeatedly occurred in the past after great ecological disasters. Stable communities are in fact exceptional in nature, and they are all but impossible wherever there is human activity. Every form of agriculture, even the most primitive, involves the creation of new ecosystems. Since most of the temperate world has been transformed by agriculture, it now constitutes a man-made ecosystem.

The conservationist's attitude toward changes in nature finds its scientific justification in the fact that it is difficult, if not impossible, to predict the long-range consequences of human interventions into natural ecosystems. Experience has shown indeed that the consequences can be disastrous. Furthermore, the progressive destruction of wilderness tends to decrease biological diversity, and this in turn renders ecological systems less stable and less likely to remain suitable for a variety of species, including man. But these scientific reasons are not likely to have much weight for many human beings. Man is adventurous and he has been endlessly taking ecological risks. His motto may well be "catastrophe rather than boredom" and, irrespective of dangers, he is likely to continue upsetting his ecosystems.

29. The most important point that the author wishes to make regarding smogs, pesticides, or strip mining is that they

(A) produce new ecosystems
(B) spoil our natural environment
(C) threaten the stability of nature
(D) have an adverse effect upon agriculture
(E) are aesthetically offensive and economically harmful

30. The author's attitude toward humanity's continued willingness to disturb ecosystems appears to be one of

(A) alarm (B) resignation (C) bewilderment (D) encouragement (E) disappointment

31. The author implies that the only scientifically defensible objection to causing environmental changes is that such changes

(A) can deprive man of important natural resources
(B) will increase the variety of biological species
(C) can encourage man to take greater ecological risks
(D) run counter to man's instinct for stability
(E) can have unpredictable consequences

32. The primary point of the passage is that

(A) most of the ecosystems in the world have been transformed by agriculture
(B) there are many differences between "natural" and "man-made" ecosystems
(C) man's nature makes most arguments to leave natural environments alone futile
(D) human beings are able to survive even great ecological disasters
(E) science has advanced man's love of adventure and his willingness to take risks

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Dylan Thomas' outward life as a poet was not, at first, in spite of his (very few) champions, easy. Following the publication of his second book, a furious attack on him developed in letters addressed to one of the two principal London newspapers. It was my privilege and pride to give the attackers more than as good as they gave. The air still seems to reverberate with the wooden sound of numskulls being soundly hit.

Although the occasion was supremely important to me, I cannot remember the exact date of our first meeting—knowing only that it was before the attack. I had reviewed his second book and had written to him; so when I arrived in London, he came to see me. It seems now, though I am so much his senior, that he and his great poetry were always a part of my life.

I have never known anyone more capable of endearing himself to others. And this was not only the result of his great charm, warmth, and touching funniness. I have never known anyone with a more holy and childlike innocence of mind. He loved humanity, and had contempt only for the cruel, the unkind (these are not always identical), and the mean. He was most generous in his enthusiasms and most loyal in his friendships. Alas, that some of the people who crowded round him were unworthy of that noble nature. But these I will leave to their shame. For he is dead. And there is nothing to be done.

33. "The wooden sound of numskulls being soundly hit" (lines 8-9) refers to the

(A) narrator's attack on those who criticized Dylan Thomas' work
(B) critics' reviews of Dylan Thomas' second book
(C) important occasion of the narrator's first meeting with Dylan Thomas
(D) holy and childlike innocence of Dylan Thomas
(E) unworthiness of some of the people who surrounded Dylan Thomas

34. It can be assumed from the passage that the narrator's review of Dylan Thomas' second book was

(A) favorable (B) explosive (C) charming (D) funny (E) angry

35. According to the passage, Dylan Thomas scorned those who lacked

(A) courage (B) humility (C) compassion (D) artistic sensitivity (E) a sense of humor

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Like the redwood which thinks the marigold may dry up its roots, Americans in the 1920's sealed their borders to most foreigners. It is ironic that the exception was the English, against whose laws we once rebelled. We were proud of our vast country, but prouder still of ourselves. The oppressed became oppressors.

It is a commentary on our institutions that it took 144 years to impose racial restrictions on a broad scale. Only in the last 44 years have attempts been made to remove them. Now that we are mending the "melting pot," we can admit that the pot had a serious crack which drained America of talent and dissipated a source of energy. The defect in the melting pot has been the restrictive immigration laws. These had their origins in sparks of xenophobia set in the 1850's and rekindled by xenophobic lobbies of the twenties. The laws were based on the national origins theory: each country was assigned a quota based on a ratio between that country's immigrants already here and the total population of the United States. Supporters of the system justified it by noting that it mirrored our society: to each country an allowance according to its prior contribution of people. But the mirror was flawed: Asians and Afro-Americans were not reflected at all.

During the presidential campaign of 1856, Lincoln said, "If the Know-Nothings obtain control, the Constitution will read: 'All men are created equal except Negroes, foreigners, and Catholics.' I shall prefer emigrating to some country where they make no pretense of loving liberty."

Not only was there an unrealistic quota system, but also problems concerning aid to needy immigrants. After the Civil War, certain states began to aid them on a small scale. To this end, the New York Board of Emigration collected a $1.50-per-passenger fee from shipowners transporting aliens. In lieu of that fee, the owner of the vessel could pay a bond on only those immigrants deemed "likely to become public charges." But any such fee did, in fact, threaten the shipowners' property rights. Thus, the Supreme Court ruled that since Congress had exclusive power to regulate foreign commerce, the states must desist. The administration of immigrant charities fell, therefore, into private hands with predictable results.

36. According to the author, a major flaw in the national origins theory was that

(A) it halted the flow of immigration
(B) it threatened property rights
(C) it created a homogeneous society
(D) certain groups were not included
(E) talented people were forced to emigrate

37. The passage is primarily concerned with

(A) explaining the background of a new law
(B) deploring a historical phenomenon
(C) formulating a theory about popular prejudice
(D) expressing doubts about the strength of liberal attitudes
(E) suggesting a revision of immigration laws

38. The author states that advocates of the quota system defended it on the grounds that it

(A) protected the fundamental principles on which the United States was founded
(B) fostered friendly ties with Western Europe
(C) prevented overpopulation and unemployment
(D) deterred the immigration of persons who were likely to become public charges
(E) reflected the existing ethnic composition of American society

39. According to the passage, the administration of immigrant charities was turned over to private agencies because

(A) the burden of administration was too great for one state agency
(B) xenophobic lobbies wished to curtail assistance to immigrants
(C) public charities were considered suspect by the populace
(D) the flow of immigration had been checked by the quota system
(E) the state administered levy was judged unconstitutional

40. The tone of Lincoln's remarks in lines 29-31 is best described as

(A) ironic (B) patronizing (C) tentative (D) defensive (E) sentimental

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 2

Time—30 minutes

35 QUESTIONS

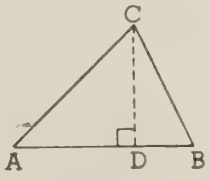
In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area = πr^2; Circumference = $2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

= is equal to	$\leqq$ is less than or equal to
$\neq$ is unequal to	$\geqq$ is greater than or equal to
$<$ is less than	$\parallel$ is parallel to
$>$ is greater than	$\perp$ is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.
If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. If $2x + 9 = -12$, then $x =$

   (A) -21 (B) $-\frac{21}{2}$ (C) $-\frac{3}{2}$ (D) 3 (E) $\frac{21}{2}$

2. $\frac{473}{3} =$

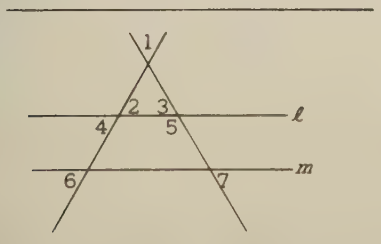
   (A) $\frac{470}{2} + \frac{3}{1}$

   (B) $\frac{400}{1} + \frac{70}{1} + \frac{3}{1}$

   (C) $\frac{473}{1} + \frac{473}{1} + \frac{473}{1}$

   (D) $\frac{473}{2} + \frac{473}{1}$

   (E) $\frac{400}{3} + \frac{70}{3} + \frac{3}{3}$



3. In the figure above, $\ell \parallel m$. Which of the following pairs of angles must be equal?

   (A) 1 and 3
   (B) 2 and 3
   (C) 3 and 5
   (D) 4 and 6
   (E) 5 and 7

4. If water flowed into a 70-liter tank and filled half the tank in 10 minutes, what was the rate of flow in liters per minute?

   (A) 3.5 (B) 7 (C) 35 (D) 70 (E) 210

5. If the perimeter of square A is double that of square B, then the area of A is how many times the area of B ?

   (A) $\frac{1}{2}$ (B) 1 (C) 2 (D) 3 (E) 4

6. What is the average of $\frac{1}{5}$ and $\frac{1}{7}$?

   (A) $\frac{1}{12}$

   (B) $\frac{1}{6}$

   (C) $\frac{6}{35}$

   (D) $\frac{12}{35}$

   (E) $\frac{36}{35}$

7. How many numbers between 1 and 99 are divisible by both 2 and 7 ?

   (A) 7 (B) 8 (C) 14 (D) 20 (E) 49

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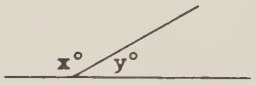
2

Questions 8-27 each consist of two quantities, one in Column A and one in Column B. You are to compare the two quantities and on the answer sheet blacken space

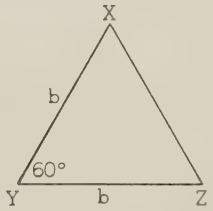
A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

Notes: 1. In certain questions, information concerning one or both of the quantities to be compared is centered above the two columns.
2. A symbol that appears in both columns represents the same thing in Column A as it does in Column B.
3. Letters such as x, n, and k stand for real numbers.

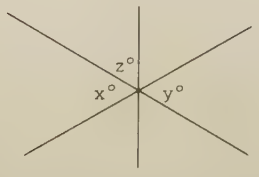
EXAMPLES

	Column A	Column B	Answers
E1.	2×6	$2 + 6$	● Ⓑ Ⓒ Ⓓ
E2.	 $180 - x$	y	Ⓐ Ⓑ ● Ⓓ
E3.	$p - q$	$q - p$	Ⓐ Ⓑ Ⓒ ●

	Column A	Column B
8.	$\frac{63{,}000{,}000}{3{,}000}$	20,000



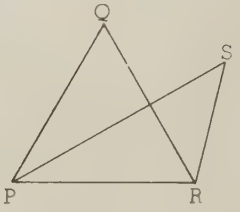
9.	Number of angles with equal measure in ΔXYZ	3
10.	$\frac{40}{9}$	4.5



Three lines intersect in a point.

	$x + z$	$y + z$

	Column A	Column B
	$a \neq 0$	
12.	$a + (-a)$	$a \cdot \frac{1}{a}$
	$k \neq 0$	
13.	$\frac{1}{k^2}$	$\left(\frac{1}{k}\right)^2$
14.	a^4b^2	a^2b^4



Perimeter of ΔPQR = Perimeter of ΔPSR

15.	PQ + QR	PS + SR

x is a positive number.

16.	x plus an increase of 50% of x	0.5x

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SUMMARY DIRECTIONS FOR COMPARISON QUESTIONS

Answer: A if the quantity in Column A is greater;
B if the quantity in Column B is greater;
C if the two quantities are equal;
D if the relationship cannot be determined from the information given.

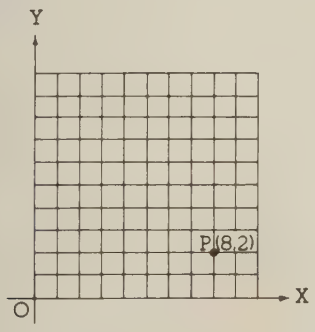
	Column A	Column B
17.	Area of a circle with radius 1	Area of a square with side 1

1 skedallion = 4.6 skippers
2 phantoms = 9.3 skippers

18.	Value of one skedallion	Value of one phantom

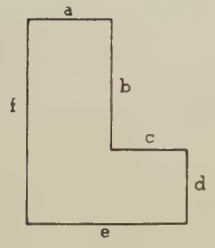
$x^2 + 8x + 15 = 0$

19.	$x^2 + 8x$	15



A point Q to be placed on the grid above has x-coordinate of 6.

20.	Distance OP	Distance OQ



In the figure above, all segments intersect at right angles and the segments have lengths a, b, c, d, e, and f as shown.

21.	$a + b + c + d$	$e + f$

	Column A	Column B
22.	The number of men in an all-man division if there are 15 patrols in the division, 14 squads in each patrol, and 4 men in each squad	The number of women in an all-woman division if there are 12 patrols in the division, 10 squads in each patrol, and 7 women in each squad

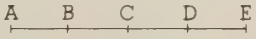
23.	$x^2 - y^2 + 2y(y + x)$	$(x + y)^2$

$-2 < x < -1$

24.	$\frac{1}{x}$	x

x and y are integers with $8 \leqq x \leqq 12$ and $2 \leqq y \leqq 10$.

25.	Maximum possible value of $x - y$	9



Note: Figure not drawn to scale.

AC = 11, BD = 7, CE = 5

26.	AD	BE

On a certain test, the average score for the juniors was 87 and the average score for the sophomores was 81.

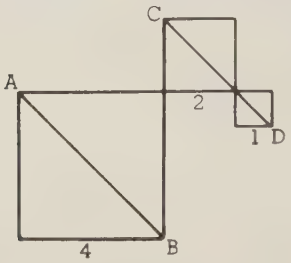
27.	The average score for the total group	84

GO ON TO THE NEXT PAGE

Solve each of the remaining problems in this section using any available space for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

28. In a certain language, a "word" is defined as any 5-letter combination in which the position of at least one letter in the "word" is in the same position in which it is found in the English alphabet. For example "dbaec" is a word, because of the placement of the letter b, but "bdaac" is not. Which of the following is a word in this language?

(A) cddcd (B) cdddc (C) ccdcc
(D) dcdcc (E) dddcd



29. In the three squares above, what is the ratio $\frac{\text{length of segment CD}}{\text{length of segment AB}}$?

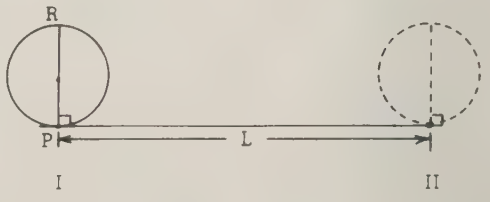
(A) $\frac{3}{16}$ (B) $\frac{3\sqrt{2}}{16}$ (C) $\frac{5}{16}$
(D) $\frac{3}{4\sqrt{2}}$ (E) $\frac{3}{4}$

30. A machine began knitting a row of 100 stitches by making 3 knit stitches and 2 purl stitches and repeated the same pattern thereafter. What is the order in which it knitted the 77th, 78th, 79th, and 80th stitches in the row?

(A) 2 knit, 2 purl
(B) 1 knit, 2 purl, 1 knit
(C) 3 knit, 1 purl
(D) 2 purl, 2 knit
(E) 1 purl, 2 knit, 1 purl

31. If a car travels x kilometers in t hours and 20 minutes, what is its average speed in kilometers per hour?

(A) $\frac{x}{t+20}$ (B) $\frac{t+20}{x}$ (C) $x\left(t+\frac{1}{3}\right)$
(D) $\frac{t+\frac{1}{3}}{x}$ (E) $\frac{x}{t+\frac{1}{3}}$



32. In the figure above, the circle at I with diameter PR is rolled one revolution along a plane until it reaches II as shown. The ratio $\frac{PR}{L}$ =

(A) $\frac{1}{\pi}$ (B) $\frac{1}{3}$ (C) 2 (D) 3 (E) π

33. The operation * is defined on ordered pairs of numbers in the following way: $(a, b) * (c, d) = (ac + bd, ad + bc)$. If $(a, b) * (x, y) = (a, b)$, then (x, y) =

(A) (0, 0)
(B) (0, 1)
(C) (1, 0)
(D) (1, 1)
(E) (1, -1)

34. For all x and y where $x \neq 2y$, $\frac{x-2y}{2y-x} + \frac{2y-x}{x-2y}$ =

(A) 2(x - 2y)
(B) 2y - x
(C) 1
(D) 0
(E) -2

35. If a person were to cut seven circles, each 4 centimeters in diameter, from a rectangular piece of cardboard 16 centimeters long, at least how many centimeters wide would the piece of cardboard have to be?

(A) $2 + 2\sqrt{3}$
(B) $4 + \sqrt{3}$
(C) $4 + 2\sqrt{3}$
(D) 6
(E) 8

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 4

Time—30 minutes

45 QUESTIONS

For each question in this section, choose the best answer and blacken the corresponding space on the answer sheet.

Each question below consists of a word in capital letters, followed by five lettered words or phrases. Choose the word or phrase that is most nearly opposite in meaning to the word in capital letters. Since some of the questions require you to distinguish fine shades of meaning, consider all the choices before deciding which is best.

Example:

GOOD: (A) sour (B) bad (C) red (D) hot (E) ugly

Ⓐ ● Ⓒ Ⓓ Ⓔ

1. COUPLE: (A) contain (B) separate (C) cover (D) increase (E) win

2. OBSOLETE: (A) indefinite (B) fertile (C) updated (D) prestigious (E) lenient

3. FLOURISH: (A) ingrain (B) wither (C) impede (D) turn pale (E) make peace

4. GRISLY: (A) rugged (B) nervous (C) modern (D) attractive (E) haughty

5. DEPENDENT: (A) self-reliant (B) intangible (C) conclusive (D) resilient (E) offensive

6. MELLOW: (A) angular (B) dry (C) clean (D) harsh (E) ragged

7. RESERVE: (A) confusion (B) admiration (C) gratitude (D) immaturity (E) boldness

8. IMPEL: (A) scatter (B) cease (C) taper off (D) turn away (E) hold back

9. PREMATURE: (A) explicit (B) perfidious (C) belated (D) punctual (E) induced

10. DILATION: (A) refraction (B) cohesion (C) vacillation (D) constriction (E) enervation

11. CONTRITE: (A) inarticulate (B) disoriented (C) unrepentant (D) infinite (E) impatient

12. MALADROIT: (A) healthful (B) zestful (C) adept (D) invigorating (E) discreet

13. IRASCIBLE: (A) imperturbable (B) curious (C) unconventional (D) unorganized (E) ordinary

14. BANE: (A) unanswerable question (B) benign influence (C) useless information (D) compilation (E) singularity

15. ERUDITION: (A) tolerance (B) crudeness (C) loyalty (D) cowardice (E) ignorance

Each sentence below has one or two blanks, each blank indicating that something has been omitted. Beneath the sentence are five lettered words or sets of words. Choose the word or set of words that best fits the meaning of the sentence as a whole.

Example:

Although its publicity has been ----, the film itself is intelligent, well-acted, handsomely produced, and altogether ----.

(A) tasteless. .respectable (B) extensive. .moderate (C) sophisticated. .amateur (D) risqué. .crude (E) perfect. .spectacular

● Ⓑ Ⓒ Ⓓ Ⓔ

16. More Americans might remain unmarried if society afforded other means of obtaining regular ----, or else valued ---- more highly.

(A) meals. .malnutrition
(B) companionship. .solitude
(C) lodging. .poverty
(D) conversation. .nonsense
(E) advice. .irresponsibility

17. The very young child has no concept of death: it is too ---- an idea, too ---- his or her experience.

(A) abstract. .removed from
(B) frightening. .thrilling for
(C) real. .close to
(D) recognizable. .irrelevant for
(E) familiar. .new to

18. Before burdening us with more of his ---- conclusions, he would do well to read a few scholarly books on the subject and brush up on the fundamentals of logic.

(A) valid (B) shrewd (C) edifying (D) superficial (E) impervious

19. An organism's energies are not called forth in some ---- fashion; rather, there are certain situations to which it invariably reacts.

(A) reasonable (B) hypothetical (C) indiscriminate (D) interacting (E) orderly

20. The book contains too much ---- material, such as extensive data about people ---- to events and unintegrated anecdotes about minor characters.

(A) erroneous. .momentous
(B) extraneous. .tangential
(C) vital. .contemporary
(D) irrelevant. .critical
(E) incontrovertible. .unattached

GO ON TO THE NEXT PAGE

4

Each passage below is followed by questions based on its content. Answer all questions following a passage on the basis of what is stated or implied in that passage.

The debilitating effects of poverty are not only the result of lack of money but are also the result of powerlessness, of being subject to one's social situation instead of being able to affect it through action, that is, through behavior that flows from an individual's decisions and plans. In other words, when social scientists have reported on the psychological consequences of poverty, it seems reasonable to believe that they have described the psychological consequences of powerlessness. The solution to poverty most frequently suggested is to help the poor secure more money without otherwise changing present power relationships. This appears to implement the idea of equality while avoiding any necessary threat to established centers of power. But, since the consequences of poverty are related to powerlessness, not to the absolute supply of money available to the poor, and since the amount of power purchasable with a given supply of money decreases as a society acquires a larger supply of goods and services, the solution of raising the incomes of the poor is likely, unless accompanied by other measures, to be ineffective in an affluent society.

In order to reduce poverty-related psychological and social problems in the United States, the major community will have to change its relationship to neighborhoods of poverty in such fashion that families in the neighborhoods have a greater stake in the broader society and can more successfully participate in the decision-making process of the surrounding community. Social action to help the poor should have the following characteristics: the poor should see themselves as the source of the action; the action should affect in major ways the preconceptions of institutions and persons defining the poor; the action should demand much in effort or skill; the action should be successful; and the successful self-originated important action should increase the feeling of potential worth and individual power of individuals who are poor.

The only initial resource which a community should provide to neighborhoods of poverty should be on a temporary basis and should consist of organizers who will enable the neighborhoods quickly to create powerful, independent, democratic organizations of the poor. Through such organizations, the poor will then negotiate with outsiders for resources and opportunities without having to submit to concurrent control from outside.

21. According to the passage, the primary role of the major community in assisting neighborhoods of poverty should be to

(A) lend experienced advice to help in the formation of democratic self-help organizations
(B) solicit public support for the business ventures of the poor
(C) provide assistance to help the poor advance socially
(D) suggest legislation to further the interests of the poor
(E) implement projects that improve conditions in the neighborhood

22. The author's main purpose is apparently to

(A) criticize present methods of helping the poor
(B) discuss various types of power and how they can be used by the poor
(C) describe the various causes of poverty
(D) propose a way in which the poor can be more effectively helped
(E) describe the psychological and social effects of power

23. The author implies that what might appear to be a lack of ambition in the poor can be ascribed to their feeling that

(A) other people succeed without working
(B) they have no friends who will support them
(C) they are unable to buy the things they want
(D) there is no one they can trust
(E) they lack control over the circumstances of their lives

24. Which of the following statements best illustrates the author's ideas about money?

(A) Self-reliance is a blessing money cannot buy.
(B) Love of money is the root of all evil.
(C) A penny saved is a penny earned.
(D) No chains enslave so well as money.
(E) Wine maketh merry but money answereth all things.

25. When speaking of power in this passage, the author is referring primarily to power as

(A) force of will
(B) authority in one's own affairs
(C) control over others
(D) energy necessary to succeed
(E) strength to resist pressure

GO ON TO THE NEXT PAGE

The quality of science that, above all else, enriches the spirits of those who pursue science is the manner in which it enables us to glimpse the orderliness of nature. One of the most thrilling experiences for the students of chemistry is their exposure to the periodic table of elements and its revelation of the systematic structure of all matter.

A comparable disclosure in astronomy grew, in large measure, out of an ambitious, laborious, and seemingly pedantic project at Harvard. Since the stars are too remote to be seen except as diffuse points of light, our knowledge of them depends heavily on their spectra. The Italian astronomer Father Angelo Secchi had already found that he could sort stars into certain spectral categories, somewhat as the first walker in a forest might pick out oaks, maples, birches, and pines. Even though the stellar types shade more into one another than tree species, Father Secchi found that they clearly fit into a succession of pigeonholes. In 1890, Harvard published a preliminary catalog of 10,351 stars, classified according to their spectra, and by 1949 the entries in the catalog had grown to 359,082.

Astronomers at Harvard soon guessed that star types reflected temperature differences, but it remained for the Indian, Meghnad Saha, to decipher the specific messages contained in the spectra of each group. The stars classed by the letter O displayed the spectral lines of highly ionized helium, silicon, and nitrogen, and thus were the hottest. The types were listed in order of decreasing temperature, as follows:

O, B, A, F, G, K, M

Then, Russell at Princeton and Hertzsprung in Europe came upon another striking pattern in star classification. If the inherent brightness, or luminosity, of the stars is plotted on a graph against their types, arranged in order of temperature, some ninety per cent of all stars lie along a narrow band that has come to be called the main sequence. The hotter a star is, if it lies on the main sequence, the brighter it is, the larger it is, and the shorter is its lifetime. The graph that displays this relationship between luminosity and star type is known as the Hertzsprung-Russell diagram. When it was formulated on the eve of the First World War, Russell viewed the main sequence as a highway along which stars travel in their long journey from birth to death. A different interpretation is now placed on the main sequence, but the Hertzsprung-Russell diagram has come to be for astronomers a testimony of order in the universe.

26. The author is primarily concerned with

(A) evaluating various schemes for classifying stars
(B) pointing out the worldwide nature of interest in astronomy
(C) explaining a basic distinction between chemistry and astronomy
(D) describing a number of projects carried on at Harvard
(E) presenting an instance of the orderly nature of the universe

27. The author compares stars with trees (lines 13-21) primarily in order to suggest that

(A) discoveries in one field may lead to new findings in another
(B) there are more classes of stars than of trees
(C) stars fall into categories that can be identified
(D) Father Secchi was primarily interested in the study of botany
(E) stars, like trees, can be classified only by experts

28. According to the passage, which of the following statements about a star classed as O is FALSE?

(A) It is likely to be brighter than a star classed as B.
(B) It is likely to be larger than a star classed as B.
(C) It is likely to be found on the main sequence.
(D) Its spectrum is likely to include lines of ionized helium, silicon, and nitrogen.
(E) It is likely to have a longer life cycle than a star classed as B.

29. Which of the following titles best summarizes the content of the passage?

(A) Measuring the Temperature of the Stars
(B) How to Determine the Luminosity of Stars
(C) The Evolution of Stars from Birth to Death
(D) The Genesis of the Hertzsprung-Russell Diagram
(E) A New Interpretation of the Hertzsprung-Russell Diagram

30. The author's tone suggests that his attitude toward the orderliness of nature is one of

(A) enthusiastic appreciation
(B) casual interest
(C) aloof disinterest
(D) anxious concern
(E) skeptical suspicion

GO ON TO THE NEXT PAGE

Select the word or set of words that best completes each of the following sentences.

31. The scientist, motivated by an abhorrence of excessively ---- explanations, usually makes a ---- attempt to reduce all phenomena to only a few known principles.

(A) convoluted. .determined
(B) simple. .futile
(C) succinct. .detailed
(D) popularized. .casual
(E) axiomatic. .redundant

32. The statesmen and politicians who found his genius ---- to their designs ---- courted his favor through four stormy decades of history.

(A) vital. .inadvertently
(B) detrimental. .languidly
(C) superfluous. .incessantly
(D) inadequate. .resolutely
(E) indispensable. .sedulously

33. This ---- law will make little difference to the ---- drug companies whose standards are already high.

(A) economical. .prosperous
(B) stringent. .scrupulous
(C) invariable. .mediocre
(D) protective. .fraudulent
(E) mitigating. .conscientious

34. Only when one approaches ---- of availability of personnel, funds, or facilities does one have to consider seriously ---- among research programs.

(A) an understanding. .disharmony
(B) an overabundance. .inadequacies
(C) a discovery. .developments
(D) a disparity. .rivalries
(E) a ceiling. .priorities

35. Modern writers, ---- to drape reality with pretty phrases, show us everything, putrid and pure, with a grim ----.

(A) aspiring. .austerity
(B) hesitating. .discretion
(C) disdaining. .objectivity
(D) purporting. .omniscience
(E) endeavoring. .naturalism

Each question below consists of a related pair of words or phrases, followed by five lettered pairs of words or phrases. Select the lettered pair that best expresses a relationship similar to that expressed in the original pair.

Example:

YAWN : BOREDOM :: (A) dream : sleep
(B) anger : madness (C) smile : amusement
(D) face : expression (E) impatience : rebellion

Ⓐ Ⓑ ● Ⓓ Ⓔ

36. SIPPING : GULPING :: (A) whispering : shouting
(B) inhaling : exhaling (C) dieting : eating
(D) inspecting : examining (E) smiling : sighing

37. OVERDUE : DEADLINE ::
(A) incorrect : error (B) surrounded : boundary
(C) accurate : measurement (D) excessive : limit
(E) luxurious : extravagance

38. GILL : SALMON :: (A) egg : chicken
(B) plume : parrot (C) lung : sparrow
(D) feather : ostrich (E) talon : eagle

39. PROPOSAL : MARRIAGE :: (A) employment : work
(B) rule : law (C) candidate : nomination
(D) truce : war (E) bid : contract

40. BIBLIOGRAPHY : PUBLICATIONS ::
(A) textbook : subject (B) museum : paintings
(C) store : goods (D) directory : names
(E) legislature : legislation

41. UNETHICAL : MORALITY ::
(A) remedial : education (B) absurd : reason
(C) exorbitant : price (D) incorrigible : freedom
(E) abstract : art

42. MIRED : MUD :: (A) soaked : skin
(B) snared : trap (C) nourished : food
(D) rescued : danger (E) extinguished : fire

43. TACITURN : SILENCE :: (A) implicated : concern
(B) suave : humility (C) compassionate : blame
(D) pompous : courtesy (E) querulous : complaint

44. MEANING : NUANCE :: (A) subtlety : nicety
(B) difference : distinction (C) form : style
(D) color : shade (E) brilliance : hue

45. DOTARD : SAGE :: (A) proxy : voter
(B) dolt : prodigy (C) student : tutor
(D) sorcerer : genius (E) mentor : counselor

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

SECTION 5

Time—30 minutes

25 QUESTIONS

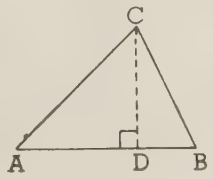
In this section solve each problem, using any available space on the page for scratchwork. Then indicate the one correct answer in the appropriate space on the answer sheet.

The following information is for your reference in solving some of the problems.

Circle of radius r: Area = πr^2; Circumference = $2\pi r$
The number of degrees of arc in a circle is 360.
The measure in degrees of a straight angle is 180.

Definitions of symbols:

=	is equal to	$\leqq$	is less than or equal to
$\neq$	is unequal to	$\geqq$	is greater than or equal to
$<$	is less than	$\parallel$	is parallel to
$>$	is greater than	$\perp$	is perpendicular to



Triangle: The sum of the measures in degrees of the angles of a triangle is 180.

If $\angle CDA$ is a right angle, then

(1) area of $\triangle ABC = \frac{AB \times CD}{2}$

(2) $AC^2 = AD^2 + DC^2$

Note: Figures which accompany problems in this test are intended to provide information useful in solving the problems. They are drawn as accurately as possible EXCEPT when it is stated in a specific problem that its figure is not drawn to scale. All figures lie in a plane unless otherwise indicated. All numbers used are real numbers.

---

1. In a certain dart game, Don scored 25 per cent more points than Craig. If Craig scored 100 points, how many points did Don score?

   (A) 225 (B) 125 (C) 100 (D) 80 (E) 75

2. For which of the following blocks could the top face viewed from the top look like the front face viewed from the front?

   (A) 

   (B) 

   (C) 

   (D) 

   (E) 

---

3. A polygon is NOT a triangle if it has exactly

   (A) three sides
   (B) three angles
   (C) one angle which measures 135°
   (D) two perpendicular sides
   (E) two parallel sides

4. $2\frac{1}{2} - \left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4} - \frac{1}{3}\right) =$

   (A) $1\frac{5}{6}$ (B) 2 (C) $2\frac{1}{12}$ (D) $2\frac{1}{6}$ (E) $2\frac{5}{12}$

5. If the freshman class has 28 students of whom $\frac{1}{2}$ are women and the sophomore class has 24 students of whom $\frac{2}{3}$ are women, which of the following gives the total number of women in both classes?

   (A) $\left(\frac{1}{2}\right) + \left(\frac{2}{3}\right)\left(\frac{24 + 28}{2}\right)$

   (B) $2\left(\frac{1}{2} + \frac{2}{3}\right)(24 + 28)$

   (C) $\left(\frac{1}{2}\right)\left(\frac{2}{3}\right)(24 + 28)$

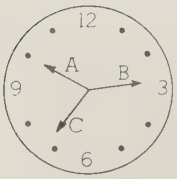
   (D) $\frac{2}{3}(28) + \frac{1}{2}(24)$

   (E) $\frac{1}{2}(28) + \frac{2}{3}(24)$

6. A roll of plastic 250 meters long costs \$26. If it takes a length of $2\frac{1}{2}$ meters of this plastic to cover a certain machine, how much will it cost to buy the exact length of plastic needed to cover 600 such machines?

   (A) \$62
   (B) \$65
   (C) \$156
   (D) \$1,550
   (E) \$1,560

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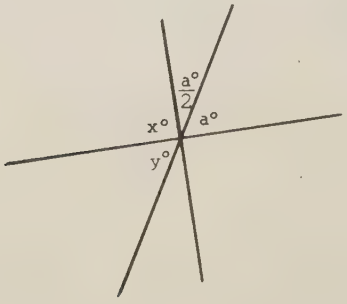
7. The clock above is accurate, but the minute, hour, and second hands are drawn the same length. If the time shown is between 2:45 and 3:00, which of the three lettered hands could be the second hand?

(A) A only
(B) B only
(C) C only
(D) Either A or C
(E) Either B or C

---

8. If the two middle digits of 4,579 are interchanged the resulting number is

(A) 18 less than 4,579
(B) 180 less than 4,579
(C) equal to 4,579
(D) 18 more than 4,579
(E) 180 more than 4,579



9. If three lines intersect in a point to form the angles shown above, then $x + y =$

(A) $\frac{a}{2} - 180$ (B) $180 - \frac{a}{2}$ (C) $180 - \frac{5a}{2}$
(D) $180 - a$ (E) $a - 180$

---

10. If $x^2 = 1$, then x^3 is equal to

(A) -3 (B) -1 only (C) 1 only
(D) 3 (E) -1 or 1

11. If $x = 2a$ and $y = \frac{1}{4a + 2}$, what is y in terms of x ?

(A) $\frac{1}{2x + 2}$

(B) $\frac{1}{2x + 4}$

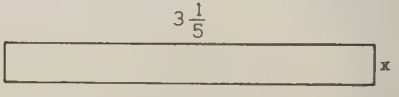
(C) $\frac{2}{x + 4}$

(D) $\frac{1}{8x + 2}$

(E) $\frac{1}{8x + 4}$

12. A certain bleach is to be diluted with water so that the ratio of bleach to water will be $\frac{1}{8}$. If 3 bottles of bleach, each containing 0.5 liters, are to be used, how many liters of water should be used to obtain this same ratio?

(A) 4
(B) 6
(C) 12
(D) 18
(E) 24



13. If the area of the rectangle above is 1, then $x =$

(A) $\frac{5}{16}$ (B) $\frac{3}{5}$ (C) $\frac{5}{3}$ (D) $\frac{16}{5}$ (E) 5

---

14. If a and b are integers and $a - b$ is a multiple of 5, which of the following must also be a multiple of 5 ?

(A) a (B) b (C) $a + b$ (D) $b - a$ (E) ab

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15. If $x > 0$, which of the following is greatest?

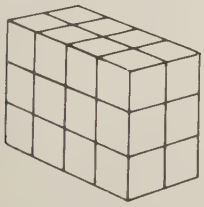
(A) $\frac{1}{x}$

(B) $\sqrt{x}$

(C) x

(D) x^2

(E) It cannot be determined from the information given.



16. In the figure above, if an edge of each small cube has length 2, what is the volume of the entire rectangular solid?

(A) 192 (B) 144 (C) 72 (D) 52 (E) 48

---

17. If 5 per cent of 50 per cent of x is 75, then $x =$

(A) 250 (B) 300 (C) 750
(D) 2,500 (E) 3,000

18. In a class of 39 students, the number b of boys is twice the number g of girls. Which of the following accurately describes this situation?

(A) $\begin{cases} b + 2g = 39 \\ b - g = 39 \end{cases}$

(B) $\begin{cases} b - 2g = 0 \\ b + g = 39 \end{cases}$

(C) $\begin{cases} 2b - g = 0 \\ b + g = 39 \end{cases}$

(D) $\begin{cases} b + g = 39 \\ b + 2g = 0 \end{cases}$

(E) $\begin{cases} b + 2g = 39 \\ b - 2g = 0 \end{cases}$

$$s = \frac{3}{2}r$$

$$r = \frac{1}{2}w$$

$$w = 3z$$

$$x = 2y$$

$$y = \frac{3}{4}z$$

19. Given the table above, x is equal to which of the following?

(A) r
(B) s
(C) w
(D) y
(E) z

---

20. If a and b are positive integers such that $\frac{a}{b} < 1$, which of the following is (are) true?

I. $a < b$

II. $\frac{1}{b} < \frac{1}{a}$

III. $\frac{a - 1}{b} < 0$

(A) None (B) I only (C) III only
(D) I and II only (E) I, II, and III

21. A teacher gave a test to 30 students and the average score was x. Scores on the test ranged from 0 to 90, inclusive. If the average score for the first 10 papers graded was 60, what is the difference between the greatest and least possible values of x ?

(A) 20 (B) 30 (C) 40 (D) 50 (E) 60

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$$x^2 + xy^2 + 2y^3 = k$$
$$y^2 + yx^2 + 2x^3 = m$$

22. In the equations above, x is an odd integer and y is an even integer. Which of the following expressions must be even?

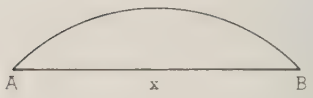
I. $m^2 + mk^2 + 2k^3$
II. $3k^3 + 3m^3 + 3k^3$
III. $k^2 + km^2 + 2m^3$

(A) I only
(B) II only
(C) III only
(D) I and II only
(E) I and III only

---

23. A triangle with one side 5 and another side 10 has perimeter P. What are the least and greatest possible integer values of P ?

(A) 21 and 29
(B) 21 and 24
(C) 16 and 39
(D) 16 and 24
(E) 6 and 29



24. If arc AB is $\frac{1}{4}$ of the circumference of a certain circle and if the length of chord AB is x, what is the diameter of the circle?

(A) $\frac{\sqrt{2}}{2}x$

(B) $\frac{\pi}{4}x$

(C) $\sqrt{2}x$

(D) $\frac{\pi}{2}x$

(E) $2\sqrt{2}x$

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25. On a 300-kilometer trip, a motorcyclist drove the first 100 kilometers at 40 kilometers per hour, the second 100 kilometers at 50 kilometers per hour, and the third 100 kilometers at 60 kilometers per hour. For what fraction of the time spent driving the 300 kilometers did the cyclist drive at 40 kilometers per hour?

(A) $\frac{4}{15}$ (B) $\frac{1}{3}$ (C) $\frac{2}{5}$ (D) $\frac{15}{37}$ (E) $\frac{11}{15}$

S T O P

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS SECTION ONLY.
DO NOT WORK ON ANY OTHER SECTION IN THE TEST.

Correct Answers for Scholastic Aptitude Test
Form Code 8G071

VERBAL		MATHEMATICAL	
Section 1	Section 4	Section 2	Section 5
1. C	1. B	1. B	1. B
2. A	2. C	2. E	2. C
3. E	3. B	3. D	3. E
4. B	4. D	4. A	4. C
5. A	5. A	5. E	5. E
6. D	6. D	6. C	6. C
7. B	7. E	7. A	7. C
8. D	8. E	*8. A	8. E
9. E	9. C	*9. C	9. B
10. E	10. D	*10. B	10. E
11. C	11. C	*11. C	11. A
12. B	12. C	*12. B	12. C
13. E	13. A	*13. C	13. A
14. D	14. B	*14. D	14. D
15. A	15. E	*15. C	15. E
16. B	16. B	*16. A	16. A
17. E	17. A	*17. A	17. E
18. A	18. D	*18. B	18. B
19. A	19. C	*19. B	19. A
20. C	20. B	*20. D	20. D
21. D	21. A	*21. C	21. E
22. B	22. D	*22. C	22. D
23. A	23. E	*23. C	23. A
24. B	24. A	*24. A	24. C
25. A	25. B	*25. A	25. D
26. D	26. E	*26. D	
27. E	27. C	*27. D	
28. C	28. E	28. B	
29. A	29. D	29. E	
30. B	30. A	30. A	
31. E	31. A	31. E	
32. C	32. E	32. A	
33. A	33. B	33. C	
34. A	34. E	34. E	
35. C	35. C	35. C	
36. D	36. A		
37. B	37. D		
38. E	38. C		
39. E	39. E		
40. A	40. D		
	41. B		
	42. B		
	43. E		
	44. D		
	45. B		

*Indicates four-choice questions. (All of the other questions are five-choice.)

The Scoring Process

Machine-scoring is done in three steps:

- *Scanning.* Your answer sheet is "read" by a scanning machine and the oval you filled in for each question is recorded on a computer tape.
- *Scoring.* The computer compares the oval filled in for each question with the correct response. Each correct answer receives one point; omitted questions do not count toward your score. For each wrong answer, a fraction of a point is subtracted to correct for random guessing. For questions with five answer choices, one-fourth of a point is subtracted for each wrong response; for questions with four answer choices, one-third of a point is subtracted for each wrong response. For example, the SAT-verbal test has 85 questions with five answer choices each, and if a student has 44 right, 32 wrong, and 9 omitted, the resulting raw score is determined as follows:

  $$44 \text{ right} - \frac{32 \text{ wrong}}{4} = 44 - 8 = 36 \text{ raw score points}$$

  Obtaining raw scores frequently involves the rounding of fractional numbers to the nearest whole number. For example, a raw score of 36.25 is rounded to 36, the nearest whole number. A raw score of 36.50 is rounded upward to 37.
- *Converting to reported scaled score.* Raw test scores are then placed on the College Board scale of 200 to 800 through a process that adjusts scores to account for minor differences in difficulty between different editions of the test. This process, known as equating, is performed so that a student's reported score is not affected by the edition of the test taken nor by the abilities of the group with whom the student takes the test. As a result of placing ATP scores on the College Board scale, scores earned by students at different times can be compared. For example, an SAT-verbal score of 400 on a test taken at one administration indicates the same level of developed verbal ability as a 400 score obtained on a different edition of the test taken at another time.

How to Score the Test

You can verify the College Board SAT scores reported to you recently by using the information in this booklet along with the copy of your answer sheet. *Before you begin, check that the form code you marked in item 3 on your answer sheet is the same as the one printed on the front of this booklet.* Compare the responses shown on the copy of your answer sheet with the list of correct answers.

SAT-Verbal Sections 1 and 4

Step A: Count the number of correct answers for *section 1* and record the number in the space provided on the worksheet on the opposite page. Then do the same for the incorrect answers. (Do not count omitted answers.) To determine subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for *section 4* and record the numbers in the spaces provided on the worksheet. To determine subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: To obtain C, add subtotal A to subtotal B, keeping any decimals.

Step D: To obtain D, your raw verbal score, round C to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.)

Step E: To find your reported SAT-verbal score, look up the total raw verbal score you obtained in step D in the conversion table on the back cover.

SAT-Mathematical Sections 2 and 5

Step A: Count the number of correct answers and the number of incorrect answers for the *four-choice questions (questions 8 through 27) in section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal A, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{3} = \text{subtotal A}$$

Step B: Count the number of correct answers and the number of incorrect answers for the *five-choice questions (questions 1 through 7 and 28 through 35) in section 2* and record the numbers in the spaces provided on the worksheet. To determine the subtotal B, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal B}$$

Step C: Count the number of correct answers and the number of incorrect answers for section 5 and record the numbers in the spaces provided on the worksheet. To determine the subtotal C, use the formula:

$$\text{number correct} - \frac{\text{number incorrect}}{4} = \text{subtotal C}$$

Step D: To obtain D, add subtotal A, subtotal B, and subtotal C, keeping any decimals.

Step E: To obtain E, your raw mathematical score, round D to the nearest whole number. (For example, any number from 44.50 to 45.49 rounds to 45.)

Step F: To find your reported SAT-mathematical score, look up the total raw mathematical score you obtained in E in the conversion table on the back cover.

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SAT-Verbal Worksheet

A. Section 1: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal A

B. Section 4: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Total unrounded raw score (Total A + B) _____ C

D. Total rounded raw score (Rounded to nearest whole number) _____ D

E. Reported SAT-verbal score (See the conversion table on the back cover.) [ ] SAT-Verbal

SAT-Mathematical Worksheet

A. Section 2: Questions 8 through 27 (4-choice) _____ (no. correct) − ⅓ (_____) (no. incorrect) = _____ subtotal A

B. Section 2: Questions 1 through 7 and 28 through 35 (5-choice) _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal B

C. Section 5: _____ (no. correct) − ¼ (_____) (no. incorrect) = _____ subtotal C

D. Total unrounded raw score (Total A + B + C) _____ D

E. Total rounded raw score (Rounded to nearest whole number) _____ E

F. Reported SAT-mathematical score (See the conversion table on the back cover.) [ ] SAT-Math

If you have any questions on these scoring instructions, you may call the phone number below.

If, after following the above scoring directions and checking your work carefully, your results disagree with the verbal or mathematical score reported on your ATP Student Report, you may request a hand scoring of your answer sheet by writing or calling:

College Board ATP
Box 592
Princeton, NJ 08541
Attention: Rescore Request
Telephone: (609) 883-8500

Please indicate whether it is your verbal or mathematical score, or both, that you want to be rescored. When you write, please include a copy of this scoring worksheet on which you did your calculations.

Score Conversion Table
Scholastic Aptitude Test
Form Code 8G071

Raw Score	College Board Reported Score		Raw Score	College Board Reported Score	
	SAT-Verbal	SAT-Math		SAT-Verbal	SAT-Math
85	800		40	450	600
84	780		39	450	590
83	770		38	440	580
82	760		37	430	570
81	750		36	430	570
80	740		35	420	560
79	720		34	410	550
78	710		33	410	540
77	700		32	400	530
76	690		31	400	520
75	680		30	390	520
74	670		29	380	510
73	660		28	380	500
72	660		27	370	490
71	650		26	360	480
70	640		25	360	470
69	640		24	350	470
68	630		23	340	460
67	630		22	340	450
66	620		21	330	440
65	610		20	330	430
64	610		19	320	420
63	600		18	310	420
62	590		17	310	410
61	590		16	300	400
60	580	800	15	290	390
59	570	780	14	290	380
58	570	770	13	280	370
57	560	760	12	270	370
56	560	750	11	270	360
55	550	740	10	260	350
54	540	730	9	260	340
53	540	720	8	250	330
52	530	710	7	240	320
51	520	700	6	240	320
50	520	690	5	230	310
49	510	680	4	220	300
48	500	670	3	220	290
47	500	660	2	210	280
46	490	650	1	200	270
45	490	640	0	200	270
44	480	630	– 1	200	260
43	470	620	– 2	200	250
42	470	620	– 3	200	240
41	460	610	– 4	200	230
			– 5	200	220
			– 6	200	220
			– 7	200	210
			– 8 or below	200	200